

August, 2026

INDIVIDUAL WORK

Question 1: A company has been allocated the network 192.168.10.0/24. The company wants to divide this network into 8 equal-sized subnets.

Required:

1. Determine the new subnet mask.
2. Determine the number of usable host addresses per subnet.
3. List the 8 subnet network addresses.
4. For each subnet, identify the first usable IP address, last usable IP address, and broadcast address.

Question 2: An organization has been assigned the network 172.16.0.0/16 and needs at least 16 subnets.

Determine:

1. The new subnet mask/prefix.
2. The number of usable hosts per subnet.
3. The network address of the first four subnets.
4. The first and last usable host addresses of the first subnet.

Question 3: A university has been allocated the network 192.168.20.0/24. The network administrator needs to create subnets for the following departments:

Department	Required Hosts
Computer Science	60
Networking	30
Administration	20
Library	10
WAN Link	2

Required:

1. Use VLSM to divide the network efficiently.
2. Determine the subnet mask/prefix for each department.
3. Assign a network address to each subnet.
4. For each subnet, identify:
 - Network address
 - First usable IP address
 - Last usable IP address
 - Broadcast address

Question Four: A company has the following networks connected to a router:

- 172.16.32.0/24
- 172.16.33.0/24
- 172.16.34.0/24
- 172.16.35.0/24
- 172.16.36.0/24
- 172.16.37.0/24
- 172.16.38.0/24
- 172.16.39.0/24

The network administrator wants to advertise all these networks using one summarized route.

Required:

1. Determine the summary network address.
2. Determine the summary prefix/subnet mask.
3. Show the binary calculation used to identify the common network bits.
4. Verify that the summarized route includes all eight original networks.
5. Explain two benefits of using this summarized route instead of advertising all eight

Question 5

- 1) Compress the IPv6 address: 2001:0db8:0000:0000:0000:ff00:0042:8329
- 2) Compress the IPv6 address: fe80:0000:0000:0000:0204:61ff:fe9d:f156
- 3) Expand the compressed IPv6 address: 2001:db8::ff00:42:8329
- 4) Expand the compressed IPv6 address: fe80::1
- 5) Compress the IPv6 address: 2607: f8b0:4005:0800:0000:0000
- 6) What is IPv6 Neighbour Discovery Protocol (NDP)? Explain its purpose and describe the main types of NDP messages used in an IPv6 network
- 7) What are IPv6 migration techniques? Name and briefly explain three IPv6 migration techniques used to enable communication between IPv4 and IPv6 networks.